Linear Algebra Essentials

Vectors, matrices, eigenvalues, and the linear algebra used throughout this book.

This appendix reviews the linear algebra that appears most often in game theory. The focus is on concepts rather than proofs, with R implementations alongside each topic. For a thorough mathematical treatment, see @strang2022.

Vectors and vector operations

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers. In game theory, vectors represent strategy profiles, probability distributions over actions, and payoff allocations.

```
# Define vectors
x <- c(3, 1, 4)
y <- c(2, 7, 1)

# Scalar multiplication and addition
2 * x
```

```
#> [1] 6 2 8
```

```
x + y
```

```
#> [1] 5 8 5
```

```
# Dot (inner) product
sum(x * y)
```

```
#> [1] 17
```

```
# Or equivalently:
x %*% y
```

```
#>      [,1]
#> [1,]   17
```

```
# Euclidean norm (length)
sqrt(sum(x^2))
```

```
#> [1] 5.09902
```

The **dot product** $\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i$ appears whenever we compute expected payoffs: if $\mathbf{p}$ is a probability vector and $\mathbf{u}$ is a payoff vector, then $\mathbf{p} \cdot \mathbf{u}$ is the expected payoff.

Matrices

A matrix $A \in \mathbb{R}^{m \times n}$ is a rectangular array of numbers. In game theory, the primary use is the payoff matrix of a normal-form game.

Matrix arithmetic in R

```r
A <- matrix(c(3, 0, 5, 1), nrow = 2, byrow = TRUE)
B <- matrix(c(3, 5, 0, 1), nrow = 2, byrow = TRUE)

# Element-wise operations
A + B
```

```
#>      [,1] [,2]
#> [1,]    6    5
#> [2,]    5    2
```

```r
A * B  # Hadamard (element-wise) product
```

```
#>      [,1] [,2]
#> [1,]    9    0
#> [2,]    0    1
```

```r
# Matrix multiplication
A %*% B
```

```
#>      [,1] [,2]
#> [1,]    9   15
#> [2,]   15   26
```

```r
# Transpose
t(A)
```

```
#>      [,1] [,2]
#> [1,]    3    5
#> [2,]    0    1
```

Key matrix operations

```r
M <- matrix(c(4, 2, 1, 3), nrow = 2, byrow = TRUE)

# Determinant
det(M)
```

```
#> [1] 10
```

```r
# Inverse (only if det != 0)
solve(M)
```

```
#>      [,1] [,2]
#> [1,]  0.3 -0.2
#> [2,] -0.1  0.4
```

```r
# Verify: M %*% M^{-1} = I
M %*% solve(M)
```

```
#>               [,1] [,2]
#> [1,]  1.000000e+00    0
#> [2,] -2.775558e-17    1
```

Systems of linear equations

Many game-theoretic computations reduce to solving a system $A\mathbf{x} = \mathbf{b}$. For example, finding a mixed Nash equilibrium in a 2x2 game requires solving the indifference equations (see @ref(sec-mixed-strategies)).

Solving $A\mathbf{x} = \mathbf{b}$

```r
# System: 2x + y = 5, x + 3y = 7
A <- matrix(c(2, 1, 1, 3), nrow = 2, byrow = TRUE)
b <- c(5, 7)

x <- solve(A, b)
x
```

```
#> [1] 1.6 1.8
```

```r
# Verify
A %*% x
```

```
#>      [,1]
#> [1,]    5
#> [2,]    7
```

Application: mixed Nash equilibrium

In a 2x2 game, finding the mixed Nash equilibrium reduces to making each player indifferent between their pure strategies. If Player 2 mixes with probability q on the first action, Player 1's indifference condition is:

$$a_{11}q + a_{12}(1-q) = a_{21}q + a_{22}(1-q)$$

This is a single linear equation in q. For larger games, mixed equilibria require solving systems of such equations simultaneously.

```r
# Prisoner's Dilemma payoffs for Player 1
# Row player payoffs: A[i,j]
A <- matrix(c(3, 0, 5, 1), nrow = 2, byrow = TRUE)

# Indifference: A[1,1]*q + A[1,2]*(1-q) = A[2,1]*q + A[2,2]*(1-q)
# Rearranging: q*(A[1,1] - A[1,2] - A[2,1] + A[2,2]) = A[2,2] - A[1,2]
denom <- A[1, 1] - A[1, 2] - A[2, 1] + A[2, 2]
q_star <- (A[2, 2] - A[1, 2]) / denom
q_star
```

```
#> [1] -1
```

Eigenvalues and eigenvectors

An eigenvector $\mathbf{v}$ of a square matrix A satisfies $A\mathbf{v} = \lambda\mathbf{v}$ for some scalar λ (the eigenvalue). Eigenvalues appear in several places in this book:

- **Replicator dynamics** (@ref(sec-replicator-dynamics)): the stability of fixed points is determined by the eigenvalues of the Jacobian matrix.
- **Markov chains**: the stationary distribution is the eigenvector corresponding to eigenvalue 1.

```
M <- matrix(c(4, 1, 2, 3), nrow = 2, byrow = TRUE)

eig <- eigen(M)
eig$values
```

```
#> [1] 5 2
```

```
eig$vectors
```

```
#>           [,1]       [,2]
#> [1,] 0.7071068 -0.4472136
#> [2,] 0.7071068  0.8944272
```

```
# Verify: M v = lambda v
lambda_1 <- eig$values[1]
v_1 <- eig$vectors[, 1]
M %*% v_1
```

```
#>          [,1]
#> [1,] 3.535534
#> [2,] 3.535534
```

```
lambda_1 * v_1
```

```
#> [1] 3.535534 3.535534
```

Stability and the Jacobian

For a dynamical system $\dot{\mathbf{x}} = f(\mathbf{x})$, a fixed point $\mathbf{x}^*$ is **stable** if all eigenvalues of the Jacobian matrix $J = \partial f/\partial \mathbf{x}$ evaluated at $\mathbf{x}^*$ have negative real parts.

```
# Example: Jacobian at a fixed point of a 2D system
J <- matrix(c(-2, 1, 0, -3), nrow = 2, byrow = TRUE)

eig_J <- eigen(J)
eig_J$values
```

```
#> [1] -3 -2
```

```r
# Both eigenvalues are negative => stable fixed point
all(Re(eig_J$values) < 0)
```

```
#> [1] TRUE
```

Positive definite matrices

A symmetric matrix A is **positive definite** if $\mathbf{x}^T A\mathbf{x} > 0$ for all nonzero $\mathbf{x}$. Equivalently, all eigenvalues are positive. Positive definiteness arises in quadratic payoff functions and in verifying second-order conditions for optimisation problems.

```r
A <- matrix(c(4, 1, 1, 3), nrow = 2, byrow = TRUE)

# Check via eigenvalues
eigen(A)$values
```

```
#> [1] 4.618034 2.381966
```

```r
all(eigen(A)$values > 0)
```

```
#> [1] TRUE
```

```r
# Check via Cholesky decomposition (succeeds only for positive definite matrices)
chol(A)
```

```
#>      [,1]     [,2]
#> [1,]    2 0.500000
#> [2,]    0 1.658312
```

Convex sets and the simplex

The probability simplex

A **mixed strategy** is a probability distribution over pure strategies. The set of all such distributions forms the **(probability) simplex**:

$$\Delta_n = \left\{ \mathbf{p} \in \mathbb{R}^n \;\middle|\; p_i \geq 0,\ \sum_{i=1}^{n} p_i = 1 \right\}$$

For $n = 2$, the simplex is a line segment; for $n = 3$, it is a triangle.

```r
# Check whether a vector is a valid mixed strategy
is_mixed_strategy <- function(p, tol = 1e-10) {
  all(p >= -tol) && abs(sum(p) - 1) < tol
}

is_mixed_strategy(c(0.3, 0.5, 0.2))
```

```
#> [1] TRUE
```

```r
is_mixed_strategy(c(0.5, 0.6, -0.1))
```

```
#> [1] FALSE
```

Convex sets and convex combinations

A set C is **convex** if for any two points $\mathbf{x}, \mathbf{y} \in C$ and any $\lambda \in [0, 1]$, the point $\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}$ is also in C. Important convex sets in game theory include:

- The simplex Δ_n (the set of mixed strategies).
- The **core** of a cooperative game (see @ref(sec-cooperative-gt)).
- The **feasible payoff region** of a repeated game (see @ref(sec-repeated-games)).

```r
# Convex combination of two strategy vectors
p1 <- c(0.7, 0.2, 0.1)
p2 <- c(0.1, 0.5, 0.4)
lambda <- 0.6

p_mix <- lambda * p1 + (1 - lambda) * p2
p_mix
```

```
#> [1] 0.46 0.32 0.22
```

```r
is_mixed_strategy(p_mix)
```

```
#> [1] TRUE
```

The support of a mixed strategy

The **support** of a mixed strategy $\mathbf{p}$ is the set of pure strategies that receive positive probability: $\text{supp}(\mathbf{p}) = \{i : p_i > 0\}$. In a Nash equilibrium, every pure strategy in the support must yield the same expected payoff (the indifference principle from @ref(sec-mixed-strategies)).

```r
support <- function(p, tol = 1e-10) {
  which(p > tol)
}

support(c(0.5, 0, 0.5, 0))
```

```
#> [1] 1 3
```

```r
support(c(0.25, 0.25, 0.25, 0.25))
```

```
#> [1] 1 2 3 4
```

Matrix decompositions

LU decomposition

R does not expose LU decomposition directly in base, but `solve()` uses it internally. For most game-theoretic applications, `solve()` suffices.

Singular value decomposition (SVD)

The SVD decomposes any $m \times n$ matrix as $A = U\Sigma V^T$. It is useful for low-rank approximations and for diagnosing near-singular payoff matrices.

```
A <- matrix(c(1, 2, 3, 4, 5, 6), nrow = 2, byrow = TRUE)

s <- svd(A)
s$d  # singular values
```

```
#> [1] 9.5080320 0.7728696
```

```
# Reconstruct: U %*% diag(d) %*% t(V) = A
s$u %*% diag(s$d) %*% t(s$v)
```

```
#>      [,1] [,2] [,3]
#> [1,]    1    2    3
#> [2,]    4    5    6
```

Summary of R linear algebra functions

Table 1: R functions for linear algebra

Task	R function	Example
Matrix multiply	`%*%`	`A %*% B`
Transpose	`t()`	`t(A)`
Determinant	`det()`	`det(A)`
Inverse	`solve()`	`solve(A)`
Solve $Ax = b$	`solve()`	`solve(A, b)`
Eigenvalues	`eigen()`	`eigen(A)$values`
SVD	`svd()`	`svd(A)$d`
Cholesky	`chol()`	`chol(A)`
Cross product	`crossprod()`	`crossprod(A)` gives A^TA
Outer product	`outer()` or `%o%`	`x %o% y`